

Address Trace Module Specification

Ver 0.1

DM270

Department: IMAGING

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HISTORY

Version	Date	Author	Notes
Ver 0.0	05/10/02	Toshiki Suzuki	Initial Release for DM270
Ver 0.1	06/17/02	Toshiki Suzuki	Added description of ATMRSV

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1. ABSTRACT

ATMモジュールはARMの命令ADDRESSをメモリにストアし、プログラム開発時のデバッグ時に使用される。専用の入出力ピンをもたず、CCSのメモリウィンドウの使用を前提としている。

2. REFERENCES

ARM7TDMI datasheet ARM DDI 0029E, Aug. 1996, ARM Ltd.

3. GLOSSARY

ATM	Address Trace Module
ARM	ARM7TDMI
AIM	ARM Internal Memory

4. FUNCTION

このATMでは、以下をサポートする：

- ◆ ARMの命令アドレスをストアし、CCSのメモリウィンドウ上のATMレジスタエリアに最新のアドレスから順に表示
- ◆ アドレスは32ビット表示と下位16ビット表示を選択できる。後者を使用することにより最大62アドレスを表示する
- ◆ レジスタをセットすることにより、アドレスのストアを停止・再開ができる

4.1 usage

- (1) リセット時、ATMの機能は有効になっており "a_lock"ビット(アドレス0x30D00/bit0)を'1'にセットするまで ARMのアドレスをストアし続ける。ユーザーが必要な場所で"a_lock"ビットを'1'にセットする。
- (2) 表示方法は2種類ありデフォルトでは32ビット幅のアドレスが連続する2つのアドレスに表示される。具体的には、ARMアドレスの履歴は、最新のものがレジスタ0x30D04と0x30D06、その前が0x30D08と0x30D0A ...のように順に表示される。次のテーブルは外部フラッシュ(0x100000)にプログラムをおいて動作させてすぐに"a_lock"ビットを'1'にセットした場合の例(Armアドレスの履歴は、最新のものから0x10002C, 0x100028, 0x100024 ...)。表示可能なアドレスは過去31アドレス。

Table 1 : register display image (default)

Address	Data
0x30D04	0x0010
0x30D06	0x002C
0x30D08	0x0010
0x30D0A	0x0028
0x30D0C	0x0010
0x30D0E	0x0024

- (3) プログラムがおかれている場所が外部メモリまたはAIMに限定されていて上位16ビットが不変の時(下位16ビットを表示すれば足りる場合)は"low16"ビット(アドレス0x30D00/bit8)を'1'にセットし上位16ビットの表示を省略可能。以下のテーブルは(2)と同様の例で表示可能なアドレスは過去62アドレス。

Table 2 : register display image ("low16"bit = 1)

Address	Data
0x30D04	0x002C
0x30D06	0x0028
0x30D08	0x0024
0x30D0A	0x0020
0x30D0C	0x001C
0x30D0E	0x0018

5. REGISTERS

Table 3 : ATM Register List

Register Name	Offset address	Description
ATMCTRL	0x00	Stop/restart storing ARM Address / Display mode ctrl
ATMRSV	0x01	Reserved register for TEST/ECO usage. Do not disclose.
ATM00	0x02	low16=0: Latest ARM instruction Address (upper16bit) low16=1: Latest ARM instruction Address (lower16bit)
ATM01	0x03	low16=0: Latest ARM instruction Address (lower16bit) low16=1: 2 nd Latest ARM instruction Address (lower16bit)
ATM02	0x04	low16=0: 2 nd Latest ARM instruction Address (upper16bit) low16=1: 3 rd Latest ARM instruction Address (lower16bit)
ATM03	0x05	low16=0: 2 nd Latest ARM instruction Address (lower16bit) low16=1: 4 th Latest ARM instruction Address (lower16bit)
ATM04	0x06	low16=0: 3 rd Latest ARM instruction Address (upper16bit) low16=1: 5 th Latest ARM instruction Address (lower16bit)
ATM05	0x07	low16=0: 3 rd Latest ARM instruction Address (lower16bit) low16=1: 6 th Latest ARM instruction Address (lower16bit)
ATM06	0x08	low16=0: 4 th Latest ARM instruction Address (upper16bit) low16=1: 7 th Latest ARM instruction Address (lower16bit)
ATM07	0x09	low16=0: 4 th Latest ARM instruction Address (lower16bit) low16=1: 8 th Latest ARM instruction Address (lower16bit)
ATM08	0x0A	low16=0: 5 th Latest ARM instruction Address (upper16bit) low16=1: 9 th Latest ARM instruction Address (lower16bit)
ATM09	0x0B	low16=0: 5 th Latest ARM instruction Address (lower16bit) low16=1: 10 th Latest ARM instruction Address (lower16bit)
ATM10	0x0C	low16=0: 6 th Latest ARM instruction Address (upper16bit) low16=1: 11 th Latest ARM instruction Address (lower16bit)
ATM11	0x0D	low16=0: 6 th Latest ARM instruction Address (lower16bit) low16=1: 12 th Latest ARM instruction Address (lower16bit)
ATM12	0x0E	low16=0: 7 th Latest ARM instruction Address (upper16bit) low16=1: 13 th Latest ARM instruction Address (lower16bit)
ATM13	0x0F	low16=0: 7 th Latest ARM instruction Address (lower16bit) low16=1: 14 th Latest ARM instruction Address (lower16bit)
ATM14	0x10	low16=0: 8 th Latest ARM instruction Address (upper16bit) low16=1: 15 th Latest ARM instruction Address (lower16bit)
ATM15	0x11	low16=0: 8 th Latest ARM instruction Address (lower16bit) low16=1: 16 th Latest ARM instruction Address (lower16bit)
ATM16	0x12	low16=0: 9 th Latest ARM instruction Address (upper16bit) low16=1: 17 th Latest ARM instruction Address (lower16bit)
ATM17	0x13	low16=0: 9 th Latest ARM instruction Address (lower16bit) low16=1: 18 th Latest ARM instruction Address (lower16bit)
ATM18	0x14	low16=0: 10 th Latest ARM instruction Address (upper16bit) low16=1: 19 th Latest ARM instruction Address (lower16bit)
ATM19	0x15	low16=0: 10 th Latest ARM instruction Address (lower16bit) low16=1: 20 th Latest ARM instruction Address (lower16bit)
ATM20	0x16	low16=0: 11 th Latest ARM instruction Address (upper16bit) low16=1: 21 st Latest ARM instruction Address (lower16bit)
ATM21	0x17	low16=0: 11 th Latest ARM instruction Address (lower16bit) low16=1: 22 nd Latest ARM instruction Address (lower16bit)
ATM22	0x18	low16=0: 12 th Latest ARM instruction Address (upper16bit) low16=1: 23 rd Latest ARM instruction Address (lower16bit)

Register Name	Offset address	Description
ATM23	0x19	low16=0: 12 th Latest ARM instruction Address (lower16bit) low16=1: 24 th Latest ARM instruction Address (lower16bit)
ATM24	0x1A	low16=0: 13 th Latest ARM instruction Address (upper16bit) low16=1: 25 th Latest ARM instruction Address (lower16bit)
ATM25	0x1B	low16=0: 13 th Latest ARM instruction Address (lower16bit) low16=1: 26 th Latest ARM instruction Address (lower16bit)
ATM26	0x1C	low16=0: 14 th Latest ARM instruction Address (upper16bit) low16=1: 27 th Latest ARM instruction Address (lower16bit)
ATM27	0x1D	low16=0: 14 th Latest ARM instruction Address (lower16bit) low16=1: 28 th Latest ARM instruction Address (lower16bit)
ATM28	0x1E	low16=0: 15 th Latest ARM instruction Address (upper16bit) low16=1: 29 th Latest ARM instruction Address (lower16bit)
ATM29	0x1F	low16=0: 15 th Latest ARM instruction Address (lower16bit) low16=1: 30 th Latest ARM instruction Address (lower16bit)
ATM30	0x20	low16=0: 16 th Latest ARM instruction Address (upper16bit) low16=1: 31 st Latest ARM instruction Address (lower16bit)
ATM31	0x21	low16=0: 16 th Latest ARM instruction Address (lower16bit) low16=1: 32 nd Latest ARM instruction Address (lower16bit)
ATM32	0x22	low16=0: 17 th Latest ARM instruction Address (upper16bit) low16=1: 33 rd Latest ARM instruction Address (lower16bit)
ATM33	0x23	low16=0: 17 th Latest ARM instruction Address (lower16bit) low16=1: 34 th Latest ARM instruction Address (lower16bit)
ATM34	0x24	low16=0: 18 th Latest ARM instruction Address (upper16bit) low16=1: 35 th Latest ARM instruction Address (lower16bit)
ATM35	0x25	low16=0: 18 th Latest ARM instruction Address (lower16bit) low16=1: 36 th Latest ARM instruction Address (lower16bit)
ATM36	0x26	low16=0: 19 th Latest ARM instruction Address (upper16bit) low16=1: 37 th Latest ARM instruction Address (lower16bit)
ATM37	0x27	low16=0: 19 th Latest ARM instruction Address (lower16bit) low16=1: 38 th Latest ARM instruction Address (lower16bit)
ATM38	0x28	low16=0: 20 th Latest ARM instruction Address (upper16bit) low16=1: 39 th Latest ARM instruction Address (lower16bit)
ATM39	0x29	low16=0: 20 th Latest ARM instruction Address (lower16bit) low16=1: 40 th Latest ARM instruction Address (lower16bit)
ATM40	0x2A	low16=0: 21 st Latest ARM instruction Address (upper16bit) low16=1: 41 st Latest ARM instruction Address (lower16bit)
ATM41	0x2B	low16=0: 21 st Latest ARM instruction Address (lower16bit) low16=1: 42 nd Latest ARM instruction Address (lower16bit)
ATM42	0x2C	low16=0: 22 nd Latest ARM instruction Address (upper16bit) low16=1: 43 rd Latest ARM instruction Address (lower16bit)
ATM43	0x2D	low16=0: 22 nd Latest ARM instruction Address (lower16bit) low16=1: 44 th Latest ARM instruction Address (lower16bit)
ATM44	0x2E	low16=0: 23 rd Latest ARM instruction Address (upper16bit) low16=1: 45 th Latest ARM instruction Address (lower16bit)
ATM45	0x2F	low16=0: 23 rd Latest ARM instruction Address (lower16bit) low16=1: 46 th Latest ARM instruction Address (lower16bit)
ATM46	0x30	low16=0: 24 th Latest ARM instruction Address (upper16bit) low16=1: 47 th Latest ARM instruction Address (lower16bit)
ATM47	0x31	low16=0: 24 th Latest ARM instruction Address (lower16bit) low16=1: 48 th Latest ARM instruction Address (lower16bit)
ATM48	0x32	low16=0: 25 th Latest ARM instruction Address (upper16bit) low16=1: 49 th Latest ARM instruction Address (lower16bit)
ATM49	0x33	low16=0: 25 th Latest ARM instruction Address (lower16bit) low16=1: 50 th Latest ARM instruction Address (lower16bit)
ATM50	0x34	low16=0: 26 th Latest ARM instruction Address (upper16bit) low16=1: 51 st Latest ARM instruction Address (lower16bit)

Register Name	Offset address	Description
ATM51	0x35	low16=0: 26 th Latest ARM instruction Address (lower16bit) low16=1: 52 nd Latest ARM instruction Address (lower16bit)
ATM52	0x36	low16=0: 27 th Latest ARM instruction Address (upper16bit) low16=1: 53 rd Latest ARM instruction Address (lower16bit)
ATM53	0x37	low16=0: 27 th Latest ARM instruction Address (lower16bit) low16=1: 54 th Latest ARM instruction Address (lower16bit)
ATM54	0x38	low16=0: 28 th Latest ARM instruction Address (upper16bit) low16=1: 55 th Latest ARM instruction Address (lower16bit)
ATM55	0x39	low16=0: 28 th Latest ARM instruction Address (lower16bit) low16=1: 56 th Latest ARM instruction Address (lower16bit)
ATM56	0x3A	low16=0: 29 th Latest ARM instruction Address (upper16bit) low16=1: 57 th Latest ARM instruction Address (lower16bit)
ATM57	0x3B	low16=0: 29 th Latest ARM instruction Address (lower16bit) low16=1: 58 th Latest ARM instruction Address (lower16bit)
ATM58	0x3C	low16=0: 30 th Latest ARM instruction Address (upper16bit) low16=1: 59 th Latest ARM instruction Address (lower16bit)
ATM59	0x3D	low16=0: 30 th Latest ARM instruction Address (lower16bit) low16=1: 60 th Latest ARM instruction Address (lower16bit)
ATM60	0x3E	low16=0: 31 st Latest ARM instruction Address (upper16bit) low16=1: 61 st Latest ARM instruction Address (lower16bit)
ATM61	0x3F	low16=0: 31 st Latest ARM instruction Address (lower16bit) low16=1: 62 nd Latest ARM instruction Address (lower16bit)

ATMCTRL (0003:0D00) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	Low16	RSV	RSV	RSV	RSV	RSV	RSV	RSV	a_lock
R/W	-	-	-	-	-	-	-	R/W	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	0	-	-	-	-	-	-	-	0

Bits	Register Name	Description
0	a_lock	0 : stores current ARM address into register 1 : stop storing current ARM address
8	low16	0 : default / 32bit ARM address 1 : only lower 16bit of the stored ARM address is displayed

ATMRSV (0003:0D02) Offset : 0x01 Do not disclose

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)	-(*)
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

(*)... R/W when ATMCTRL bit4 is set '1'

Bits	Register Name	Description
15-0	--- (RSV)	Reserved register for TEST/ECO usage. Available when ATMCTRL bit4 is set '1'.

ATM00-ATM61 (0003:0D04-0003:0D7E) Offset : 0x02-0x3F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ATMn[15:0] (n=00,01,...,61)															
R/W	R															
Default	0x0000															

Bits	Register Name	Description
15:0	ATMn (n=00,01,..61)	ARM address history

6. Interface

Table 4 : ATM Top-level I/O List

Pin Name	Bits	I/O	Fr/To	Description
Functional clocking / common signals				
rst_n	1	I	ext	Power on reset..
clk_arm	1	I	clkc	ARM clock.
ARM interface signals				
arm_a	32	I	ARM	Address bus from ARM.
arm_nmreq	1	I	ARM	Memory request from ARM.
arm_nopc	1	I	ARM	'not op-code fetch' signal from ARM
BUSC interface signals				
Busc_a	6	I	busc	Address input bus.
Busc_do	16	I	busc	Data input bus.
Busc_nrw	1	I	busc	0:read / 1:write.
Busc_ncs_atm	1	I	busc	Chip select. Active low.
Busc_nwait	1	I	busc	Active low wait
atm_do	16	O	busc	Data output bus.
Test				
tst_scan	1	I	tst	Scan test enable

7. TEST Methodology / Others

- When ATMCTRL bit4 is set '1', offset 0x01 become a writable register. It's a reserved registers for ECO use.